

# Shahriar Khan

## Computer Vision & Deep Learning Researcher

Dhaka, Bangladesh | +880 1975-759760 | [turjo410@gmail.com](mailto:turjo410@gmail.com)  
[github.com/turjo410](https://github.com/turjo410) | [linkedin.com/in/shahriar-khan-08770b289](https://linkedin.com/in/shahriar-khan-08770b289) | [skp.engineer](https://skp.engineer) | [kaggle.com/turjo410](https://kaggle.com/turjo410)

**Research Interests:** Computer Vision | Video Understanding | Explainable AI | Efficient / Edge Deep Learning

## Education

**B.Sc. in Computer Science & Engineering**, East West University, Dhaka

2022–2026

CGPA: **3.92/4.00** | TOEFL iBT: 93

Undergraduate Thesis: *TrafficGuard AI* (Supervisor: Dr. Sarwar Jahan)

## Research Experience

### TrafficGuard AI – Multi-Camera Traffic Surveillance Platform

[GitHub](#)

*Tools:* Python, Flask, Flutter, YOLOv11, YOLO-World, OpenCV, RTSP, Socket.IO

- **Deployment:** Designed and deployed an industry-linked traffic-monitoring platform with the Dhaka Metropolitan Police (DMP) and Spectrum Engineering Consortium; formally recognized by DMP, Spectrum, and Cybernetics Hi-Tech Solutions (2026).
- **Computer vision pipeline:** Built a custom YOLOv11 concurrent RTSP pipeline for **16 localized vehicle classes** with real-time speed estimation, wrong-way, crowd, collision, and incident-review workflows.
- **Edge deployment:** Compared **6 YOLO architectures** via ablation studies to optimize real-time inference on edge devices.
- **Integrated system:** Combined detection, analytics, and incident review into a single Flutter mobile app for operators.

### Banana XAI – Explainable Image Classification

[Live](#) | [GitHub](#)

*Tools:* PyTorch, Streamlit, Grad-CAM, Grad-CAM++, Eigen-CAM, Ablation-CAM, LIME

- **Benchmark:** Benchmarked **12 CNN and vision-transformer models** for image classification and compared four explanation methods (Grad-CAM, Grad-CAM++, Eigen-CAM, LIME) in an interactive dashboard.
- **Reproducibility:** Built a side-by-side interface for reviewing predictions and explanation maps, turning model-comparison experiments into a reproducible visual analysis tool.

## Publications

- **Supervised, Semi-Supervised, and Self-Supervised Learning for Brain MRI Pathology Detection: Benchmarking YOLO with SimCLR and DINOv3.** *Engineering Reports*, Wiley, **under review**, 2026.
- **Dust Cloud Detection: A Machine Learning Approach.** *Data Mining and Information Security*, LNNS 1915, pp. 77–86, Springer, 2026. [DOI](#)
- **Data Transmission in Internet of Things Networks with Minimal Energy Consumption.** *International Journal of Engineering Research & Technology*, vol. 14, no. 05, 2026. [Article](#)

## Teaching Experience

### East West University

November 2024 – April 2026

*Undergraduate Teaching Assistant, Department of CSE*

*Dhaka, Bangladesh*

- Mentored **120+ undergraduate students** across three semesters through programming labs, code reviews, grading, and debugging in Python, C++, and Java.

### East West University Programming Club

December 2023 – June 2026

*Member and Event Volunteer* — supported programming contests and technical workshops.

## Honors & Awards

**Project recognition:** Dhaka Metropolitan Police; Spectrum Engineering Consortium; Cybernetics Hi-Tech Solutions (2026).

**Scholarships:** Full Tuition-Free Merit (2022); Merit Scholarship and Dean's List (2024); Dean's Merit (2025).

## Technical Skills

**AI / CV / ML:** PyTorch, TensorFlow, YOLOv11, YOLO-World, OpenCV, scikit-learn, Grad-CAM family

**Programming:** Python, C++, Java, SQL, LaTeX

**Research / Deliv-Git, Docker, Linux, Flask, Streamlit, REST APIs, Socket.IO**

**ery:**

**Additional:** Flutter, React, Next.js, Node.js, PostgreSQL, Firebase

## Additional Projects

### Election Analytics 2026 – ML Dashboard

[Live](#) | [GitHub](#)

*Tools:* Next.js, PySpark MLlib, React Native

- Built a PySpark MLlib classification workflow (84.78% accuracy) with choropleth and seat-map visualizations.

### PromptMaster Pro – AI Learning SaaS

[Live](#) | [GitHub](#)

*Tools:* React, TypeScript, Supabase

- Launched a learning platform serving 200+ members with 59+ lessons across 5 tracks.